A

Project Report

on

ContextBasedRAG: A Context-Aware Retrieval-Augmented Generation Assistant for Factory Maintenance

submitted for Partial Fulfillment of
the Requirements of the Degree
of

Bachelor of Technology

in

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to

G H Raisoni College of Engineering and Management, Jalgaon

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CERTIFICATE

This is to certify that the project entitled "***ContextBasedRAG: A Context-Aware Retrieval-Augmented Generation Assistant for Factory Maintenance"***, submitted by

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in partial fulfillment of the degree of *Bachelor of Technology* in *CSE (Artificial Intelligence and Machine Learning)* has been satisfactorily carried out under my guidance as per the requirement of G H Raisoni College of Engineering and Management, Jalgaon.

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Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25 (probabilistic keyword ranking function)
CMMS	Computerised Maintenance Management System
CSV	Comma-Separated Values
DFD	Data Flow Diagram
DOCX / PPTX / XLSX	Word / PowerPoint / Excel Open XML file formats
HMAC	Hash-based Message Authentication Code
HNSW	Hierarchical Navigable Small World (graph index)
HTTP(S)	HyperText Transfer Protocol (Secure)
JSON	JavaScript Object Notation
LLM	Large Language Model
LOTO	Lockout / Tagout
MTBF / MTTR	Mean Time Between Failures / Mean Time To Repair
NLP	Natural Language Processing
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RRF	Reciprocal Rank Fusion
SHA-256	Secure Hash Algorithm, 256-bit
SMTP	Simple Mail Transfer Protocol
SOP	Standard Operating Procedure
SPA	Single Page Application
UML	Unified Modeling Language
UPI	Unified Payments Interface

Nomenclature

k1	BM25 term-frequency saturation constant (1.5)
b	BM25 length-normalisation constant (0.75)
f(t, d)	Frequency of term t in chunk d
|d|	Length of chunk d in tokens
avgdl	Average chunk length of the organization
N	Number of chunks of the organization
n_t	Number of chunks that contain term t
IDF(t)	Inverse document frequency of term t
rank_i(c)	Position of chunk c in ranking i
k	Number of fused chunks kept as evidence (6)
m	Word-overlap score of two claim sentences
TP, FP, FN	True positives, false positives, false negatives

Abstract

Maintenance knowledge in small factories is scattered across equipment manuals, standard operating procedures and breakdown registers, and general chatbots answer from general knowledge and may invent unsafe values. This project presents ContextBasedRAG, a web application that answers maintenance questions only from a factory's own records. Documents are split into page-aware chunks and searched by hybrid retrieval, which merges vector search and BM25 keyword ranking through Reciprocal Rank Fusion. Google Gemini writes the answer from numbered evidence; the backend validates every citation and replies "not found" when no evidence supports an answer. A deterministic detector reports conflicting values between documents to a supervisor instead of choosing one. The system also manages assets, breakdowns, approved work orders, permissions and an audit trail. On 26 labelled questions, all labelled sources were retrieved within the top three passages and every unanswerable question was refused; 137 automated tests pass.

Keywords: Retrieval-Augmented Generation, Large Language Models, Hybrid Search, BM25, Reciprocal Rank Fusion, Vector Database, Citation Grounding, Conflict Detection, Maintenance Management, Industry 4.0.